NLP Coursework

Patronising and Condescending Language (PCL) Detection

0. Setup: Install Dependencies & Download Data

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import os

os.chdir('/content/drive/MyDrive/ColabNotebooks/nlp-pcl')
print(os.getcwd())
```

```
/content/drive/MyDrive/ColabNotebooks/nlp-pcl
```

```
# Install required libraries (wordcloud not available by default in Colab)
!pip install wordcloud -q
```

```
import os
import re
import urllib.request
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
from collections import Counter
from wordcloud import WordCloud
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.util import ngrams

# Download required NLTK resources
nltk.download('punkt', quiet=True)
nltk.download('punkt_tab', quiet=True)
nltk.download('stopwords', quiet=True)

print('All imports successful.')
```

```
All imports successful.
```

1. Load and Prepare the Dataset

```
# ----------------------------------------------------------------
# Load the main TSV file. Columns:
#   par_id, art_id, keyword, country_code, text, label (0-4 scale)
# ----------------------------------------------------------------

df = pd.read_csv(
    'dontpatronizeme_pcl.tsv',
    sep='\t',
    header=None,
    names=['par_id', 'art_id', 'keyword', 'country_code', 'text', 'label'],
    skiprows=4,
    quoting=3
)

print(f'Total paragraphs loaded: {len(df)}')
print(f'Label distribution (0-4 scale):\n{df["label"].value_counts().sort_index()}')
```

```
Total paragraphs loaded: 10469
Label distribution (0-4 scale):
label
0    8529
1     947
2     144
3     458
4     391
Name: count, dtype: int64
```

```
# ----------------------------------------------------------------
# Convert the 5-point scale to binary labels per the task spec:
#   Labels 0-1  -->  No PCL (0)
#   Labels 2-4  -->  PCL    (1)
# Then split into Train / Dev using official paragraph ID files.
# ----------------------------------------------------------------

df['binary_label'] = (df['label'] >= 2).astype(int)

def load_par_ids(filepath):
    ids = set()
    with open(filepath, 'r') as f:
        next(f)  # skip header row in the CSV
        for line in f:
            line = line.strip()
            if line:
                par_id = line.split(',')[0].strip()
                ids.add(par_id)
    return ids

train_ids = load_par_ids('train_semeval_parids-labels.csv')
dev_ids   = load_par_ids('dev_semeval_parids-labels.csv')

df['par_id'] = df['par_id'].astype(str).str.strip()

train_df = df[df['par_id'].isin(train_ids)].copy()
dev_df   = df[df['par_id'].isin(dev_ids)].copy()

print(f'Train set size : {len(train_df)}')
print(f'Dev   set size : {len(dev_df)}')
print(f'\nTrain binary label distribution:\n{train_df["binary_label"].value_counts()}')
```

```
Train set size : 8375
Dev   set size : 2094

Train binary label distribution:
binary_label
0    7581
1     794
Name: count, dtype: int64
```

Exercise 2

EDA Technique 1: Basic Statistical Profiling

```
# ----------------------------------------------------------------
# Tokenise each paragraph and compute per-token counts.
# We use NLTK's word_tokenize for consistency with downstream steps.
# ----------------------------------------------------------------

train_df['tokens']      = train_df['text'].apply(word_tokenize)
train_df['token_count'] = train_df['tokens'].apply(len)

# Summary statistics per class
stats = train_df.groupby('binary_label')['token_count'].agg(
    Mean='mean', Median='median', Min='min', Max='max', Std='std'
).round(1)
stats.index = ['No PCL (0)', 'PCL (1)']

print('Token Count Statistics by Class:')
print(stats.to_string())
```

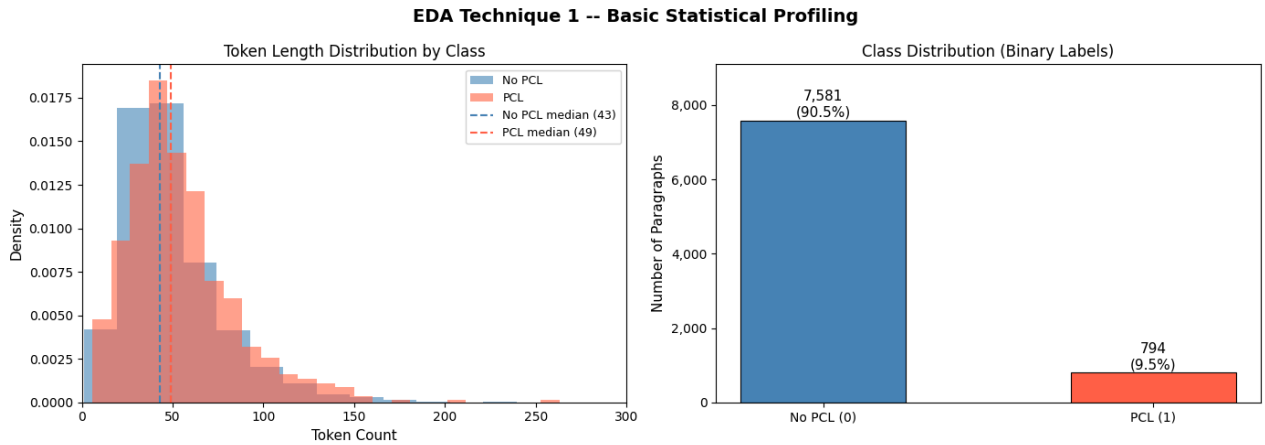
```
Token Count Statistics by Class:
            Mean  Median  Min  Max   Std
No PCL (0)  49.4    43.0    1  918  29.8
PCL (1)     55.1    49.0    6  520  34.1
```

```
# ----------------------------------------------------------------
# Vocabulary size: count unique lowercase tokens per class
# and overall across the full training set.
# ----------------------------------------------------------------

all_tokens   = [t.lower() for toks in train_df['tokens'] for t in toks]
pcl_tokens   = [t.lower() for toks in train_df[train_df['binary_label']==1]['tokens'] for t in toks]
nopcl_tokens = [t.lower() for toks in train_df[train_df['binary_label']==0]['tokens'] for t in toks]

print(f'Overall vocabulary size : {len(set(all_tokens)):,} unique tokens')
```

```
print(f'PCL vocabulary size     : {len(set(pcl_tokens)):,} unique tokens')
print(f'No-PCL vocabulary size  : {len(set(nopcl_tokens)):,} unique tokens')
```

```
Overall vocabulary size : 29,311 unique tokens
PCL vocabulary size     : 6,969 unique tokens
No-PCL vocabulary size  : 27,810 unique tokens
```

```
# ----------------------------------------------------------------
# Figure 1a: Token length distribution (histogram, both classes)
# Figure 1b: Class distribution (bar chart with counts + %)
# ----------------------------------------------------------------

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle('EDA Technique 1 -- Basic Statistical Profiling', fontsize=14, fontweight='bold')

# --- 1a: Token length distribution ---
pcl_lengths   = train_df[train_df['binary_label']==1]['token_count']
nopcl_lengths = train_df[train_df['binary_label']==0]['token_count']

axes[0].hist(nopcl_lengths, bins=50, alpha=0.6, color='steelblue', label='No PCL', density=True)
axes[0].hist(pcl_lengths,   bins=50, alpha=0.6, color='tomato',    label='PCL',    density=True)
axes[0].axvline(nopcl_lengths.median(), color='steelblue', linestyle='--', linewidth=1.5,
                label=f'No PCL median ({int(nopcl_lengths.median())})')
axes[0].axvline(pcl_lengths.median(),   color='tomato',    linestyle='--', linewidth=1.5,
                label=f'PCL median ({int(pcl_lengths.median())})')
axes[0].set_xlabel('Token Count', fontsize=11)
axes[0].set_ylabel('Density', fontsize=11)
axes[0].set_title('Token Length Distribution by Class')
axes[0].legend(fontsize=9)
axes[0].set_xlim(0, 300)

# --- 1b: Class distribution ---
class_counts = train_df['binary_label'].value_counts().sort_index()
total = class_counts.sum()
bars = axes[1].bar(
    ['No PCL (0)', 'PCL (1)'],
    class_counts.values,
    color=['steelblue', 'tomato'],
    edgecolor='black', linewidth=0.8, width=0.5
)
for bar, count in zip(bars, class_counts.values):
    axes[1].text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 30,
        f'{count:,}\n({100*count/total:.1f}%)',
        ha='center', va='bottom', fontsize=11
    )
axes[1].set_title('Class Distribution (Binary Labels)', fontsize=12)
axes[1].set_ylabel('Number of Paragraphs', fontsize=11)
axes[1].set_ylim(0, class_counts.max() * 1.2)
axes[1].yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{int(x):,}'))

plt.tight_layout()
plt.savefig('eda_technique1_statistical_profiling.png', dpi=150, bbox_inches='tight')
plt.show()
```



EDA Technique 2: Lexical Analysis

```
# ---------------------------------------------------------------
# Prepare filtered token lists.
# For n-gram and word cloud analysis we remove stop words and
# punctuation to surface content-bearing terms.
# We keep the full token lists separately for stop word density.
# ---------------------------------------------------------------

STOPWORDS = set(stopwords.words('english'))

def get_clean_tokens(token_list, min_len=2):
    """Lowercase, remove stop words and non-alphabetic tokens."""
    return [
        t.lower() for t in token_list
        if t.isalpha() and len(t) >= min_len and t.lower() not in STOPWORDS
    ]

pcl_df   = train_df[train_df['binary_label'] == 1]
nopcl_df = train_df[train_df['binary_label'] == 0]

pcl_clean   = [t for toks in pcl_df['tokens']   for t in get_clean_tokens(toks)]
nopcl_clean = [t for toks in nopcl_df['tokens'] for t in get_clean_tokens(toks)]

print(f'PCL   -- content tokens (no stops): {len(pcl_clean):,}')
print(f'No-PCL -- content tokens (no stops): {len(nopcl_clean):,}')
```

```
PCL   -- content tokens (no stops): 19,812
No-PCL -- content tokens (no stops): 177,470
```

```
# ---------------------------------------------------------------
# Stop Word Density: proportion of each paragraph's alpha tokens
# that are stop words. Computed per example, then averaged per class.
# ---------------------------------------------------------------

def stopword_density(token_list):
    alpha = [t.lower() for t in token_list if t.isalpha()]
    if not alpha:
        return 0.0
    return sum(1 for t in alpha if t in STOPWORDS) / len(alpha)

train_df['stop_density'] = train_df['tokens'].apply(stopword_density)

density_stats = train_df.groupby('binary_label')['stop_density'].agg(
    Mean='mean', Median='median', Std='std'
).round(3)
density_stats.index = ['No PCL', 'PCL']
print('Stop Word Density by Class:')
print(density_stats.to_string())
```

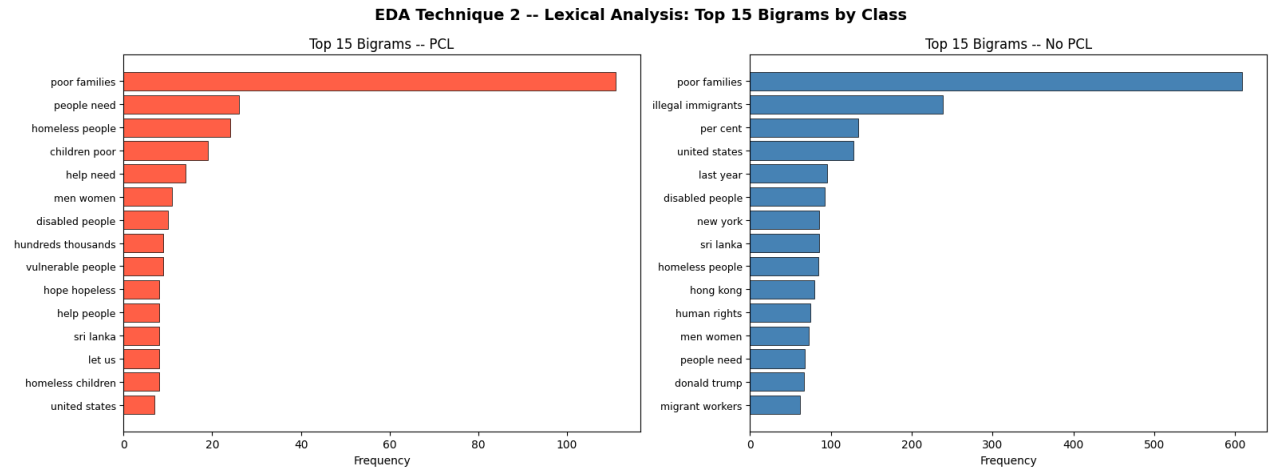
```
Stop Word Density by Class:
         Mean  Median    Std
No PCL  0.425   0.429  0.088
PCL     0.453   0.459  0.087
```

```
# ---------------------------------------------------------------
# N-gram Analysis: compute top 15 bigrams and trigrams per class.
# Using stop-word-filtered tokens to surface meaningful phrases.
# ---------------------------------------------------------------

def top_ngrams(token_list, n, top_k=15):
    ng = list(ngrams(token_list, n))
    return Counter(ng).most_common(top_k)

pcl_bigrams    = top_ngrams(pcl_clean,   2)
nopcl_bigrams  = top_ngrams(nopcl_clean, 2)
pcl_trigrams   = top_ngrams(pcl_clean,   3)
nopcl_trigrams = top_ngrams(nopcl_clean, 3)

# Print bigrams
for label, ng_list in [('PCL Bigrams', pcl_bigrams), ('No-PCL Bigrams', nopcl_bigrams)]:
    print(f'\nTop 15 {label}:')
    for ng, count in ng_list:
        print(f'  {", ".join(ng):<35} {count}')
```

```

Top 15 PCL Bigrams:
  poor, families                      111
  people, need                        26
  homeless, people                    24
  children, poor                      19
  help, need                          14
  men, women                          11
  disabled, people                    10
  hundreds, thousands                 9
```

```
  vulnerable, people                  9
  hope, hopeless                      8
  help, people                        8
  sri, lanka                          8
  let, us                             8
  homeless, children                  8
  united, states                      7

Top 15 No-PCL Bigrams:
  poor, families                      609
  illegal, immigrants                 239
  per, cent                           134
  united, states                      128
  last, year                          95
  disabled, people                    92
  new, york                           85
  sri, lanka                          85
  homeless, people                    84
  hong, kong                          80
  human, rights                       75
  men, women                          73
  people, need                        68
  donald, trump                       67
  migrant, workers                    62
```

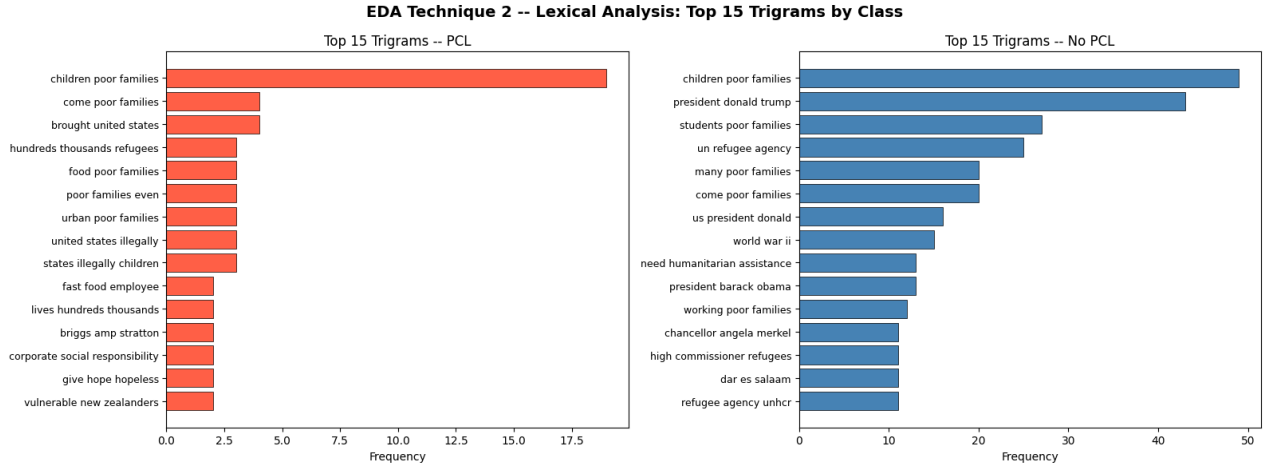
```
# ----------------------------------------------------------------
# Figure 2a: Top 15 bigrams for PCL vs No-PCL (horizontal bars)
# ----------------------------------------------------------------

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle('EDA Technique 2 -- Lexical Analysis: Top 15 Bigrams by Class',
             fontsize=14, fontweight='bold')

def plot_ngrams(ax, ngram_list, title, color):
    labels = [' '.join(ng) for ng, _ in ngram_list]
    counts = [c for _, c in ngram_list]
    ax.barh(labels[::-1], counts[::-1], color=color, edgecolor='black', linewidth=0.5)
    ax.set_title(title, fontsize=12)
    ax.set_xlabel('Frequency', fontsize=10)
    ax.tick_params(axis='y', labelsize=9)

plot_ngrams(axes[0], pcl_bigrams,   'Top 15 Bigrams -- PCL',    'tomato')
plot_ngrams(axes[1], nopcl_bigrams, 'Top 15 Bigrams -- No PCL', 'steelblue')

plt.tight_layout()
plt.savefig('eda_technique2a_bigrams.png', dpi=150, bbox_inches='tight')
plt.show()
print('Figure 2a saved.')
```




```
Figure 2a saved.
```

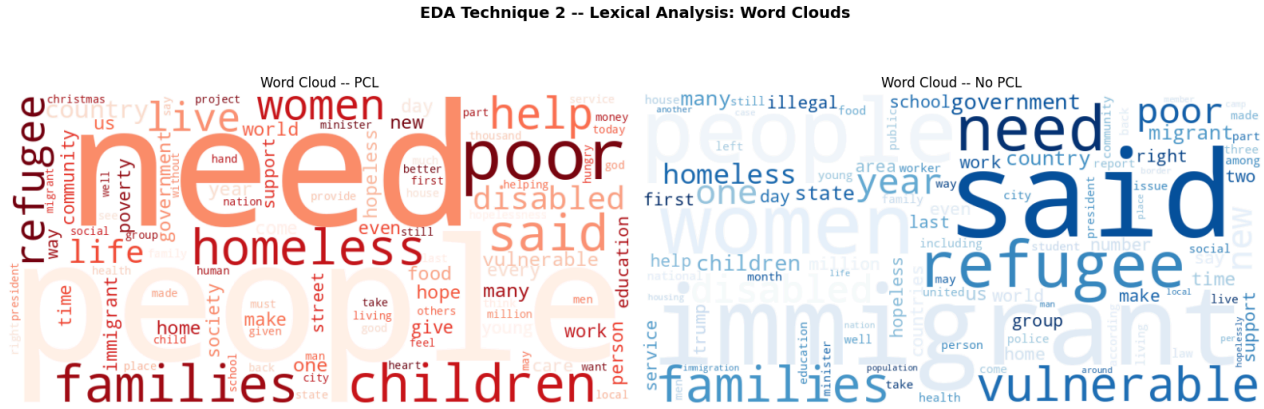
```
# ----------------------------------------------------------------
# Figure 2b: Top 15 trigrams for PCL vs No-PCL (horizontal bars)
# ----------------------------------------------------------------

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle('EDA Technique 2 -- Lexical Analysis: Top 15 Trigrams by Class',
             fontsize=14, fontweight='bold')

plot_ngrams(axes[0], pcl_trigrams,   'Top 15 Trigrams -- PCL',    'tomato')
plot_ngrams(axes[1], nopcl_trigrams, 'Top 15 Trigrams -- No PCL', 'steelblue')

plt.tight_layout()
```

```
plt.savefig('eda_technique2b_trigrams.png', dpi=150, bbox_inches='tight')
plt.show()
print('Figure 2b saved.')
```




```
Figure 2b saved.
```

```
# ----------------------------------------------------------------
# Figure 2c: Word clouds for PCL and No-PCL
# ----------------------------------------------------------------

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle('EDA Technique 2 -- Lexical Analysis: Word Clouds',
             fontsize=14, fontweight='bold')

for ax, tokens, title, colormap in [
    (axes[0], pcl_clean,   'PCL',    'Reds'),
    (axes[1], nopcl_clean, 'No PCL', 'Blues')
]:
    wc = WordCloud(
        width=800, height=400,
        background_color='white',
        colormap=colormap,
        max_words=100,
        collocations=False
    ).generate(' '.join(tokens))
    ax.imshow(wc, interpolation='bilinear')
    ax.axis('off')
    ax.set_title(f'Word Cloud -- {title}', fontsize=12)

plt.tight_layout()
plt.savefig('eda_technique2c_wordclouds.png', dpi=150, bbox_inches='tight')
plt.show()
print('Figure 2c saved.')
```

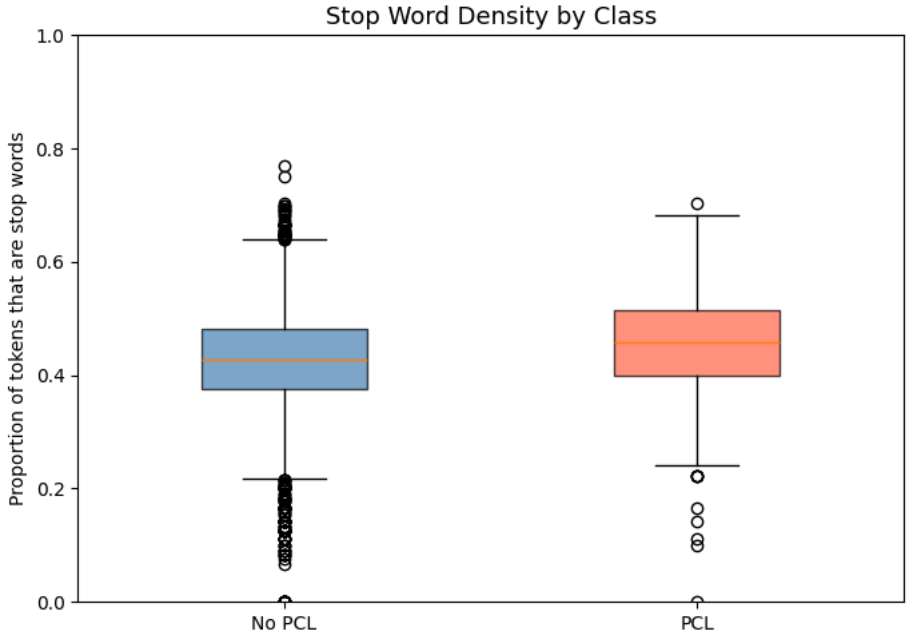



```
Figure 2c saved.
```

```
# ----------------------------------------------------------------
# Figure 2d: Stop word density box plot by class
# ----------------------------------------------------------------

fig, ax = plt.subplots(figsize=(7, 5))

data_to_plot = [
    train_df[train_df['binary_label']==0]['stop_density'].values,
```

```
    train_df[train_df['binary_label']==1]['stop_density'].values
]
bp = ax.boxplot(data_to_plot, patch_artist=True, labels=['No PCL', 'PCL'], widths=0.4)
for patch, color in zip(bp['boxes'], ['steelblue', 'tomato']):
    patch.set_facecolor(color)
    patch.set_alpha(0.7)

ax.set_title('Stop Word Density by Class', fontsize=13)
ax.set_ylabel('Proportion of tokens that are stop words', fontsize=10)
ax.set_ylim(0, 1)

plt.tight_layout()
plt.savefig('eda_technique2d_stopword_density.png', dpi=150, bbox_inches='tight')
plt.show()
print('Figure 2d saved.')
```

```
/tmp/ipykernel_870/2793232064.py:11: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tic
  bp = ax.boxplot(data_to_plot, patch_artist=True, labels=['No PCL', 'PCL'], widths=0.4)
```




```
Figure 2d saved.
```

Exercise 4: Model Training

Setup: Install and Import

```
# Install HuggingFace Transformers and related libraries
!pip install transformers datasets -q
```

```
import os
import numpy as np
import pandas as pd
import torch
from torch.utils.data import Dataset, DataLoader
from transformers import (
    RobertaTokenizerFast,
    RobertaForSequenceClassification,
    get_linear_schedule_with_warmup
)
from torch.optim import AdamW
from sklearn.metrics import f1_score, classification_report
import warnings
warnings.filterwarnings('ignore')

# Use GPU if available
DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Using device: {DEVICE}')
```

```
Using device: cuda
```

```
DATA_PATH = ''
```

```
df = pd.read_csv(
    DATA_PATH + 'dontpatronizeme_pcl.tsv',
    sep='\t',
    header=None,
    names=['par_id', 'art_id', 'keyword', 'country_code', 'text', 'label'],
    skiprows=4,
    quoting=3
)

df['label'] = df['label'].astype(int)
df['binary_label'] = (df['label'] >= 2).astype(int)
df['par_id'] = df['par_id'].astype(str).str.strip()

def load_par_ids(filepath):
    ids = set()
    with open(filepath, 'r') as f:
        next(f)  # skip header
        for line in f:
            line = line.strip()
            if line:
                ids.add(line.split(',')[0].strip())
    return ids

train_ids = load_par_ids(DATA_PATH + 'train_semeval_parids-labels.csv')
dev_ids   = load_par_ids(DATA_PATH + 'dev_semeval_parids-labels.csv')

train_df = df[df['par_id'].isin(train_ids)].reset_index(drop=True)
dev_df   = df[df['par_id'].isin(dev_ids)].reset_index(drop=True)

print(f'Train: {len(train_df)} | Dev: {len(dev_df)}')
print(f'Train PCL: {train_df["binary_label"].sum()} | Train No-PCL: {(train_df["binary_label"]==0).sum()}')
```

```
Train: 8375 | Dev: 2094
Train PCL: 794 | Train No-PCL: 7581
```

```
# pip install -q transformers datasets accelerate scikit-learn torch

import os
import numpy as np
import torch

from sklearn.metrics import f1_score
from datasets import Dataset
from transformers import (
    AutoTokenizer, AutoModelForSequenceClassification,
    TrainingArguments, Trainer, DataCollatorWithPadding, set_seed
)

set_seed(42)

# ---------------------------
# 1) Config
# ---------------------------
MODEL_NAME = "roberta-base"
OUTPUT_DIR = "./bestmodel_roberta_base"

TEXT_COL = "text"
LABEL_COL = "binary_label"

MAX_LENGTH = 192
BATCH_SIZE = 16
LR = 2e-5
EPOCHS = 4
WEIGHT_DECAY = 0.01
WARMUP_STEPS = 300  # explicit warmup

os.makedirs(OUTPUT_DIR, exist_ok=True)

# ---------------------------
# 2) Ensure labels are ints AND text is string
# ---------------------------
train_df[LABEL_COL] = train_df[LABEL_COL].astype(int)
dev_df[LABEL_COL]   = dev_df[LABEL_COL].astype(int)

# Ensure text column is string and handle potential NaNs
train_df[TEXT_COL] = train_df[TEXT_COL].fillna('').astype(str)
dev_df[TEXT_COL] = dev_df[TEXT_COL].fillna('').astype(str)

# ---------------------------
# 3) Build HF Datasets (labels must be named "labels")
# ---------------------------
train_ds = Dataset.from_pandas(train_df[[TEXT_COL, LABEL_COL]], preserve_index=False)
```

```
dev_ds   = Dataset.from_pandas(dev_df[[TEXT_COL, LABEL_COL]], preserve_index=False)

train_ds = train_ds.rename_column(LABEL_COL, "labels")
dev_ds   = dev_ds.rename_column(LABEL_COL, "labels")

tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME, use_fast=True)

def tokenize(batch):
    return tokenizer(batch[TEXT_COL], truncation=True, max_length=MAX_LENGTH)

train_ds = train_ds.map(tokenize, batched=True, remove_columns=[TEXT_COL])
dev_ds   = dev_ds.map(tokenize, batched=True, remove_columns=[TEXT_COL])

data_collator = DataCollatorWithPadding(tokenizer=tokenizer)

# ---------------------------
# 4) Model
# ---------------------------
model = AutoModelForSequenceClassification.from_pretrained(MODEL_NAME, num_labels=2)

# ---------------------------
# 5) Metrics: best-threshold F1 for positive class
# ---------------------------
THRESHOLDS = np.linspace(0.05, 0.95, 19)

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    probs = torch.softmax(torch.tensor(logits), dim=-1).numpy()[:, 1]

    # F1 at 0.5 (reference)
    preds_05 = (probs >= 0.5).astype(int)
    f1_05 = f1_score(labels, preds_05, pos_label=1)

    # Best threshold F1
    best_f1 = -1.0
    best_thr = 0.5
    for t in THRESHOLDS:
        f1 = f1_score(labels, (probs >= t).astype(int), pos_label=1)
        if f1 > best_f1:
            best_f1 = f1
            best_thr = t

    return {
        "f1_pos@0.5": float(f1_05),
        "best_f1_pos": float(best_f1),
        "best_thr": float(best_thr),
    }

# ---------------------------
# 6) Training args
#   IMPORTANT: your env uses eval_strategy, not evaluation_strategy
# ---------------------------
training_args = TrainingArguments(
    output_dir=OUTPUT_DIR,
    eval_strategy="epoch",
    save_strategy="epoch",
    load_best_model_at_end=True,

    metric_for_best_model="best_f1_pos",
    greater_is_better=True,

    learning_rate=LR,
    per_device_train_batch_size=BATCH_SIZE,
    per_device_eval_batch_size=BATCH_SIZE,
    num_train_epochs=EPOCHS,
    weight_decay=WEIGHT_DECAY,

    warmup_steps=WARMUP_STEPS,

    fp16=False,
    bf16=False,

    max_grad_norm=1.0,
    logging_steps=100,
    report_to="none",
    save_total_limit=2,
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_ds,
```

```
    eval_dataset=dev_ds,
    data_collator=data_collator,
    compute_metrics=compute_metrics,
)

trainer.train()

# ---------------------------
# 7) Final dev predictions using best threshold
# ---------------------------
pred = trainer.predict(dev_ds)
logits = pred.predictions
labels = pred.label_ids
probs = torch.softmax(torch.tensor(logits), dim=-1).numpy()[:, 1]

best_f1 = -1.0
best_thr = 0.5
for t in THRESHOLDS:
    f1 = f1_score(labels, (probs >= t).astype(int), pos_label=1)
    if f1 > best_f1:
        best_f1 = f1
        best_thr = t

print(f"FINAL Best Dev F1(pos): {best_f1:.4f} at threshold {best_thr:.2f}")

dev_preds = (probs >= best_thr).astype(int)

dev_out_path = os.path.join(OUTPUT_DIR, "dev.txt")
with open(dev_out_path, "w") as f:
    for y in dev_preds:
        f.write(f"{int(y)}\n")

print("Wrote:", dev_out_path)

# Save model + tokenizer
model_dir = os.path.join(OUTPUT_DIR, "model")
os.makedirs(model_dir, exist_ok=True)
trainer.save_model(model_dir)
tokenizer.save_pretrained(model_dir)
print("Saved model to:", model_dir)
```

```
Map: 100%                                          8375/8375 [00:02<00:00, 2868.30 examples/s]

Map: 100%                                          2094/2094 [00:00<00:00, 2504.74 examples/s]

Loading weights: 100%                              197/197 [00:00<00:00, 241.01it/s, Materializing param=roberta.encoder.layer.11.output.dense.weight]
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                              | Status     |
---------------------------------+------------+-
lm_head.bias                     | UNEXPECTED |
lm_head.layer_norm.bias          | UNEXPECTED |
lm_head.dense.bias               | UNEXPECTED |
roberta.embeddings.position_ids  | UNEXPECTED |
lm_head.dense.weight             | UNEXPECTED |
lm_head.layer_norm.weight        | UNEXPECTED |
classifier.dense.bias            | MISSING    |
classifier.out_proj.bias         | MISSING    |
classifier.dense.weight          | MISSING    |
classifier.out_proj.weight       | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

[2096/2096 15:06, Epoch 4/4]

Epoch	Training Loss	Validation Loss	F1 Pos@0.5	Best F1 Pos	Best Thr
1	0.255806	0.222288	0.549801	0.549801	0.500000
2	0.193360	0.235527	0.566893	0.576744	0.550000
3	0.120598	0.274131	0.520635	0.588529	0.050000
4	0.057355	0.407296	0.588235	0.602597	0.900000

```
Writing model shards: 100%                                    1/1 [00:02<00:00,  2.26s/it]

Writing model shards: 100%                                    1/1 [00:02<00:00,  2.14s/it]

Writing model shards: 100%                                    1/1 [00:08<00:00,  8.90s/it]

Writing model shards: 100%                                    1/1 [00:03<00:00,  3.86s/it]
There were missing keys in the checkpoint model loaded: ['roberta.embeddings.LayerNorm.weight', 'roberta.embeddings.LayerNor
There were unexpected keys in the checkpoint model loaded: ['roberta.embeddings.LayerNorm.beta', 'roberta.embeddings.LayerNo
FINAL Best Dev F1(pos): 0.6026 at threshold 0.90
Wrote: ./bestmodel_roberta_base/dev.txt
Writing model shards: 100%                                    1/1 [00:02<00:00,  2.19s/it]
Saved model to: ./bestmodel_roberta_base/model
```

```
import pandas as pd
import torch
import numpy as np
import os
from datasets import Dataset
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification,
    Trainer,
    DataCollatorWithPadding,
    TrainingArguments
)

# ---------------------------
# 1) Paths
# ---------------------------
MODEL_DIR = "./bestmodel_roberta_base/model"
OUTPUT_DIR = "./bestmodel_roberta_base"

TEST_FILE = "task4_test.tsv"
TEXT_COL = "text"
MAX_LENGTH = 192

# IMPORTANT:
# Replace with your saved threshold value from dev
best_thr = 0.9

# ---------------------------
# 2) Load model + tokenizer
# ---------------------------
tokenizer = AutoTokenizer.from_pretrained(MODEL_DIR)
model = AutoModelForSequenceClassification.from_pretrained(MODEL_DIR)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
```

```
# Minimal inference-only TrainingArguments
training_args = TrainingArguments(
    output_dir=OUTPUT_DIR,
    per_device_eval_batch_size=32,
    do_train=False,
    do_predict=True,
    logging_strategy="no",
    report_to="none"
)

data_collator = DataCollatorWithPadding(tokenizer=tokenizer)

trainer = Trainer(
    model=model,
    args=training_args,
    data_collator=data_collator,
)

# ---------------------------
# 3) Load test TSV
# ---------------------------
test_df = pd.read_csv(
    TEST_FILE,
    sep="\t",
    header=None,
    names=["par_id", "art_id", "keyword", "country_code", "text"],
    quoting=3
)

print("Test size:", len(test_df))
print(test_df.head())

assert TEXT_COL in test_df.columns, f"{TEXT_COL} column not found!"

# ---------------------------
# 4) Build Dataset
# ---------------------------
test_ds = Dataset.from_pandas(
    test_df[[TEXT_COL]],
    preserve_index=False
)

def tokenize(batch):
    return tokenizer(
        batch[TEXT_COL],
        truncation=True,
        padding="max_length",
        max_length=MAX_LENGTH
    )

test_ds = test_ds.map(
    tokenize,
    batched=True,
    remove_columns=[TEXT_COL]
)

# ---------------------------
# 5) Predict probabilities
# ---------------------------
pred = trainer.predict(test_ds)

logits = pred.predictions
probs = torch.softmax(torch.tensor(logits), dim=-1).numpy()[:, 1]

test_preds = (probs >= best_thr).astype(int)

# ---------------------------
# 6) Write test.txt
# ---------------------------
os.makedirs(OUTPUT_DIR, exist_ok=True)
test_out_path = os.path.join(OUTPUT_DIR, "test.txt")

with open(test_out_path, "w") as f:
    for y in test_preds:
        f.write(f"{int(y)}\n")

print("Wrote:", test_out_path)
print("Used threshold:", best_thr)
print("Total predictions:", len(test_preds))
```

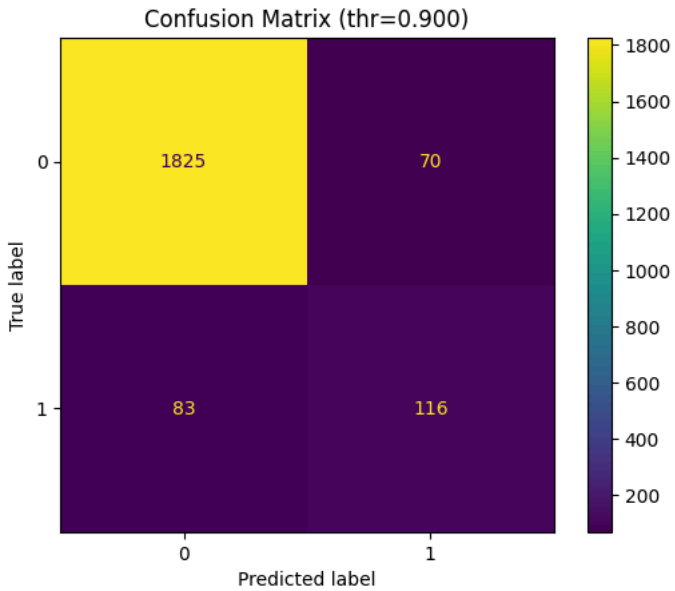
```
Loading weights: 100%                                    201/201 [00:00<00:00, 495.70it/s, Materializing param=roberta.encoder.layer.11.output.dense.weight]
Test size: 3832
  par_id      art_id     keyword country_code  \
0    t_0   @@7258997  vulnerable           us
1    t_1  @@16397324       women           pk
2    t_2  @@16257812     migrant           ca
3    t_3   @@3509652     migrant           gb
4    t_4    @@477506  vulnerable           ca

                                                text
0  In the meantime , conservatives are working to...
1  In most poor households with no education chil...
2  The real question is not whether immigration i...
3  In total , the country 's immigrant population...
4  Members of the church , which is part of Ken C...
Map: 100%                                    3832/3832 [00:01<00:00, 2220.93 examples/s]
Wrote: ./bestmodel_roberta_base/test.txt
Used threshold: 0.9
Total predictions: 3832
```

Exercise 5.2: Error Analysis and Local Evaluation

```
# ============================================================
# 1) Confusion Matrix
# ============================================================

import re
import numpy as np
import pandas as pd
import torch

from sklearn.metrics import (
    confusion_matrix,
    ConfusionMatrixDisplay,
    classification_report,
    f1_score
)
import matplotlib.pyplot as plt

TEXT_COL = "text"
LABEL_COL = "binary_label"


pred_dev = trainer.predict(dev_ds)
logits_dev = pred_dev.predictions
y_true = pred_dev.label_ids.astype(int)

p_pcl = torch.softmax(torch.tensor(logits_dev), dim=-1).numpy()[:, 1]   # P(class=1)
y_pred = (p_pcl >= best_thr).astype(int)

err = dev_df.copy().reset_index(drop=True)

if LABEL_COL in err.columns:
    err["y_true"] = err[LABEL_COL].astype(int)
else:
    err["y_true"] = y_true

err["p_pcl"] = p_pcl
err["y_pred"] = y_pred
err["correct"] = (err["y_true"].values == err["y_pred"].values)
err["margin"] = np.abs(err["p_pcl"] - best_thr)

err["n_words"] = err[TEXT_COL].astype(str).apply(lambda s: len(s.split()))

def assign_case(row):
    if row["y_true"] == 0 and row["y_pred"] == 1:
        return "FP"
    if row["y_true"] == 1 and row["y_pred"] == 0:
        return "FN"
    if row["y_true"] == 1 and row["y_pred"] == 1:
        return "TP"
    return "TN"

err["case"] = err.apply(assign_case, axis=1)

cm = confusion_matrix(err["y_true"], err["y_pred"])
print("Confusion matrix:\n", cm, "\n")
print("Classification report:\n")
print(classification_report(err["y_true"], err["y_pred"], digits=4))
```

```
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=[0, 1])
disp.plot(values_format="d")
plt.title(f"Confusion Matrix (thr={best_thr:.3f})")
plt.show()
```

```
Confusion matrix:
 [[1825   70]
 [  83  116]]

Classification report:

              precision    recall  f1-score   support

           0     0.9565    0.9631    0.9598      1895
           1     0.6237    0.5829    0.6026       199

    accuracy                         0.9269      2094
   macro avg     0.7901    0.7730    0.7812      2094
weighted avg     0.9249    0.9269    0.9258      2094
```




```
# ============================================================
# 2) False Positives Analysis
# ============================================================

def show_examples(df, case, n=10):
    if case == "FP":
        # predicted PCL with high confidence, but actually not PCL
        ex = df[df["case"]=="FP"].sort_values("p_pcl", ascending=False).head(n)
    elif case == "FN":
        # missed true PCL with low p_pcl (confidently wrong)
        ex = df[df["case"]=="FN"].sort_values("p_pcl", ascending=True).head(n)
    else:
        ex = df[df["case"]==case].head(n)

    cols = ["p_pcl", "y_true", "y_pred", "keyword", "country_code", "n_words", "text"]
    return ex[cols]

show_examples(err, "FP", n=25)
```

	p_pcl	y_true	y_pred	keyword	country_code	n_words	text
972	0.998425	0	1	refugee	us	64	In an act of defiance against Hungarian author...
1687	0.998403	0	1	in-need	za	36	"Adopt a Mission serves as a platform for the ...
1745	0.998401	0	1	poor-families	ke	44	""" We have identified extremely poor families...
416	0.998366	0	1	refugee	ke	19	"E-mail Address : * <h> A clinic called "" Hop...
1252	0.998343	0	1	poor-families	ng	72	Usually children of poor families remember the...
1140	0.998329	0	1	hopeless	nz	41	""" It beggars belief this scheme has been cob...
1557	0.998304	0	1	vulnerable	jm	26	Meanwhile , Dr. Thomas told JIS News that the ...
1510	0.998257	0	1	hopeless	jm	46	""" Is it the bad gas , the ZIKV , is it the j...
715	0.998216	0	1	in-need	my	40	Jesus begins his teaching in Matthew with the ...
254	0.998147	0	1	hopeless	au	31	""" So we do need to heal ourselves as an Abor...
1499	0.998107	0	1	hopeless	ke	139	I have been looking to see if Ojinga 's son an...
388	0.998087	0	1	in-need	bd	48	""" Your personal leadership has been critical...
1097	0.998080	0	1	homeless	lk	144	A happy day it was indeed when a 31 homeless s...
1781	0.998077	0	1	homeless	ke	28	In other words , Waitiki 's now settled squatt...
1064	0.998070	0	1	hopeless	ph	66	"But more than taking personal strength from t...
1130	0.998020	0	1	in-need	in	82	Christmas is celebration of the birth of not m...
1040	0.998011	0	1	hopeless	bd	46	The majority of these people now reside , in r...
660	0.998009	0	1	homeless	ph	16	The homeless pitch in whatever money they have...
585	0.997936	0	1	in-need	ph	84	""" I always consider this job as a gift , bei...
1119	0.997927	0	1	in-need	ph	18	""" The boxing community is wonderful at helpi...
1542	0.997870	0	1	homeless	jm	7	Columns <h> Poor , pregnant and homeless
747	0.997802	0	1	hopeless	ph	18	ATD wants to uplift the self-esteem of MNC res...
854	0.997653	0	1	vulnerable	ng	35	In another development , Eiremiokhae was also ...
872	0.997640	0	1	poor-families	my	29	Earlier at the ceremony , Azlan , who is also ...
373	0.997511	0	1	in-need	ph	35	Dacawi practically pioneered the concept of ci...

```
# ============================================================
# 3) False Negatives Analysis
# ============================================================

show_examples(err, "FN", n=25)
```

	p_pcl	y_true	y_pred	keyword	country_code	n_words	text
150	0.000216	1	0	disabled	hk	32	Cheung said 20 disabled undergraduate students...
121	0.000264	1	0	poor-families	us	37	The World Health Organization did not give a r...
35	0.000269	1	0	women	za	76	"Mari ? tte Coetzee from Stofberg Family Viney...
163	0.000282	1	0	refugee	ng	16	Many refugees do n't want to be resettled anyw...
464	0.000352	1	0	poor-families	pk	45	A massive cleanup and awareness drive was cond...
52	0.000353	1	0	immigrant	gh	74	"Nearly 15,000 West African teenagers leave th...
108	0.000387	1	0	hopeless	bd	44	"Calling for an immediate political solution t...
42	0.000424	1	0	hopeless	pk	47	The Global Gender Gap Report 2016 has ranked P...
12	0.000459	1	0	homeless	ie	30	It 's calculated that over 204,000 days of pur...
155	0.000496	1	0	poor-families	ng	38	""" Nigeria 's tertiary education system is st...
1966	0.000508	1	0	poor-families	hk	70	A top health official said today that the gove...
41	0.000526	1	0	vulnerable	ng	51	"He said : "" I think we can consider introduc...
54	0.000541	1	0	migrant	pk	30	Famously progressive San Francisco has among A...
129	0.000596	1	0	in-need	ph	31	"After the satisfying evening , Manny Osme ? a...
142	0.000601	1	0	poor-families	au	48	Michael Gove 's recent suggestion that inadequ...
114	0.000625	1	0	hopeless	lk	93	The indigenous Palestinians see themselves aba...
95	0.000684	1	0	disabled	nz	97	New Zealand has been getting more comfortable ...
386	0.000688	1	0	poor-families	ca	62	"I am shocked , disgusted and dismayed at yet ...
0	0.000690	1	0	homeless	ke	75	"His present "" chambers "" may be quite humbl...
124	0.000755	1	0	hopeless	lk	80	Some years later the daughter 's marriage brok...
24	0.000817	1	0	homeless	ie	19	Mum living in homeless shelter has ' nowhere t...
101	0.000857	1	0	poor-families	ca	79	"So when journalists and observers tell us tha...
106	0.000895	1	0	homeless	tz	35	He said the victims who are currently rendered...
44	0.000981	1	0	poor-families	tz	27	t is remiss not to mention here that not all s...
48	0.000989	1	0	migrant	hk	33	The measures have kept the migrants living in ...

```
# ============================================================
# 4) Group substitution
# ============================================================

import re
import numpy as np
import pandas as pd
import torch
from datasets import Dataset

# Define suitable substitution pairs
SUB_PAIRS = [
    ("women", "men"),
    ("immigrant", "citizen"),
    ("migrant", "local resident"),
    ("refugee", "citizen"),
    ("homeless", "housed"),
    ("disabled", "able-bodied"),
    ("poor-families", "wealthy families"),
    ("in-need", "well-off"),
]

def _surface_forms(keyword: str):
    """
    Generate surface forms to match in text.
    Handles hyphenated keywords by adding a spaced variant.
    Example: "poor-families" -> ["poor-families", "poor families"]
    """
    forms = [keyword]
    if "-" in keyword:
        forms.append(keyword.replace("-", " "))
    seen = set()
    out = []
    for f in forms:
        if f not in seen:
            out.append(f)
            seen.add(f)
    return out
```

```
    return out

def substitute_group(text: str, src_keyword: str, tgt_phrase: str) -> str:
    """
    Replace occurrences of src_keyword (and its surface variants) with tgt_phrase
    using case-insensitive, word-boundary matching.
    """
    s = str(text)
    for src_form in _surface_forms(src_keyword):
        pattern = re.compile(r"\b" + re.escape(src_form) + r"\b", flags=re.IGNORECASE)
        s = pattern.sub(tgt_phrase, s)
    return s

def predict_p1_texts(texts, batch_size=32):
    """
    Predict P(class=1) for raw texts using the existing `trainer`+`tokenizer`.
    """
    tmp_df = pd.DataFrame({TEXT_COL: list(texts)})
    tmp_ds = Dataset.from_pandas(tmp_df, preserve_index=False)

    def tok(batch):
        return tokenizer(batch[TEXT_COL], truncation=True, padding=True)

    tmp_ds = tmp_ds.map(tok, batched=True, remove_columns=[TEXT_COL])
    out = trainer.predict(tmp_ds)
    p1 = torch.softmax(torch.tensor(out.predictions), dim=-1).numpy()[:, 1]
    return p1

# Run substitution analysis
sub_rows = []

for src, tgt in SUB_PAIRS:
    # Focus on rows where the example's keyword equals the source keyword
    # (so the substitution is relevant/meaningful).
    mask = err["keyword"].astype(str).str.lower() == src.lower()
    sub = err[mask].copy()
    if len(sub) == 0:
        continue

    sub["src"] = src
    sub["tgt"] = tgt
    sub["text_cf"] = sub[TEXT_COL].apply(lambda t: substitute_group(t, src, tgt))

    # Keep only rows where something actually changed
    changed = sub["text_cf"] != sub[TEXT_COL].astype(str)
    sub = sub[changed].copy()
    if len(sub) == 0:
        continue

    # Predict counterfactual probabilities
    sub["p_cf"] = predict_p1_texts(sub["text_cf"].tolist())
    sub["delta_p"] = sub["p_cf"] - sub["p_pcl"]

    sub["y_pred_cf"] = (sub["p_cf"] >= best_thr).astype(int)
    sub["flip"] = (sub["y_pred_cf"] != sub["y_pred"]).astype(int)

    sub_rows.append(sub)

if len(sub_rows) == 0:
    print("No substitution results: no rows matched or no text changes occurred.")
else:
    sub_all = pd.concat(sub_rows, ignore_index=True)

    # Summary per substitution pair
    summary = (
        sub_all.groupby(["src", "tgt"], dropna=False)
        .agg(
            n=("src", "size"),
            avg_p_orig=("p_pcl", "mean"),
            avg_p_cf=("p_cf", "mean"),
            avg_delta_p=("delta_p", "mean"),
            med_delta_p=("delta_p", "median"),
            flip_rate=("flip", "mean"),
        )
        .sort_values(["flip_rate", "avg_delta_p"], ascending=[False, False])
        .reset_index()
    )

    display(summary)

    # Show the most sensitive examples overall
    sub_all["abs_delta"] = sub_all["delta_p"].abs()
    most_sensitive = sub_all.sort_values("abs_delta", ascending=False).head(20)
```